Optimal Transport

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1 Introduction to Optimal Transport

1.1 Monge's formulation:

Definition 1. ***Transport map.*** *Given two probability spaces, X and Y, and associated measures $\mu \in P(X)$ and $\nu \in P(Y)$ a transport map, $T : X \to Y$, is a function such that*

$$\nu(B) = \mu(T^{-1}(B)) \tag{1}$$

for all ν-measurable sets B. We say that T transports μ to ν, denoted by $\nu = T_{\#}\mu$.

Remark. *In general, if T is a Y-measurable function from $X \to Y$, then the function $f_{\#} : \Sigma_Y \to [0, \infty]$ defined by $f_{\#}(B) := \mu(T^{-1}(B)) \ \ \forall B \in \Sigma_Y$ is a measure. This measure is often called the pushforward measure*

One major consequence of this definition is that for a given two probability spaces, there may not be any transport maps between them. To see this, let $\mu = \delta_{x_1}$ and $\nu = \frac{1}{2}\delta_{y_1} + \frac{1}{2}\delta_{y_2}$. If T is a transport map from μ to ν, then $\nu(\{y_1\}) = \frac{1}{2}$, but the only possibilities forμ are $\mu(T^{-1}\{y_1\}) = 0$ or $\mu(T^{-1}\{y_1\}) = 1$.

Below is a proposition that is useful for calculating pushforward measures

Proposition 1. *Given a measure, μ, and its pushforward measure, $T_{\#}\mu$ the change of variables formula for a pushforward measure is:*

$$\int_Y f(y)d(T_{\#}\mu)(y) = \int_X f(T(x))d\mu(x) \tag{2}$$

Definition 2. ***Monge's Optimal Transport Problem*** *Given a cost function, $c : X \times Y \to \mathbb{R}$, $\mu \in P(X)$, and $\nu \in P(Y)$, then the Monge's Optimal Transport Problem is,*

$$\min \mathbb{M}(T) := \int c(x, T(x))d\mu(x) \qquad \text{subject to } \nu = T_{\#}\mu \tag{3}$$

1.2 Kantorovich's formulation:

Definition 3. ***Coupling*** *A coupling of measures μ and ν, is a measure, $\pi \in P(X \times Y)$, such that $\pi(A \times Y) = \mu(A)$ and $\pi(X \times B) = \nu(B)$ for all $A \subseteq X, B \subseteq Y$*

The condition of $\pi(A \times Y) = \mu(A)$ and $\pi(X \times B) = \nu(B)$ can also be called as the marginals. The set of all couplings is called the set of transport plans. We denote this by $\Pi(\mu, \nu)$. One useful consequence of this definition is that $\mu \times \nu \in \Pi(\mu, \nu) \ \ \forall \mu, \nu$. Thus $\Pi(\mu, \nu)$ is never empty, which will be useful later. The advantage of this definition over the definition of transport maps is that a coupling allows for the mass of a discrete point in a measure to be split. For example, $\mu = \delta_{x_1}$ and $\nu = \frac{1}{2}\delta_{y_1} + \frac{1}{2}\delta_{y_2}$ does have a transport plan between them.

Definition 4. ***Kantorovich's Optimal Transport Problem*** *Let $\mu \in P(X)$ and $\nu \in P(Y)$, then the Kantorovich's Optimal Transport Problem is*

$$\min_{\pi \in \Pi(\mu,\nu)} \mathbb{K}(\pi) := \int_{X \times Y} c(x, y)d\pi(x, y) \tag{4}$$

2 Special Case: Discrete Measures

The two formulations of Optimal transport are reduced to linear optimization problems when working with only discrete measures. This is an important result since any probability measure can be approximated by a discrete measure. In addition, linear optimization is a well studied topic and there are efficient algorithms to solve linear optimization problems. In the discrete case, transport maps and couplings are linear functions, and thus can be represented by a matrix. The cost function can also be represented by a matrix by defining the matrix C by $C_{ij} = \|x_i - x_j\|^2$.

2.1 Equal Mass Discrete Measures

One of the more practical results from Optimal Transport theory is the special case when μ and ν are both a specific discrete measures. For this result, we can let our cost function be any cost function. Suppose that $\mu = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}$ and $\nu = \frac{1}{n}\sum_{i=1}^{n} \delta_{y_i}$. What makes these discrete measures special, is that each point has equal mass and their are the same number of points with mass in the μ measure and the ν measure. This is sufficient for the Monge formulation to exist. Theorem 2.6 and 2.7 from [3] shows that these measures has a solution for the Monge's formulation and further more, the solution will be a permutation matrix, i.e., $T(x_i) = y_{\sigma(i)}$ for some permutation map $\sigma : \{1, 2, 3, ..., n\} \to \{1, 2, 3, ..., n\}$.

2.1.1 Example: Color Transfer

One example of discrete optimal transport is the color transfer between two images. Suppose we have two pictures with $x_i = (r_i, g_i, b_i) \in [0, 1]^3$ be the RGB value of the ith pixel of the first picture and $y_i = (r_i, g_i, b_i) \in [0, 1]^3$ be the RGB value of the ith pixel of the second picture. Then using optimal transport, we can create an optimal transport between the color values of the first picture and color values of the second picture, which in simple terms is optimally swapping the color palettes of the two pictures.



Public Domain images sourced from Raw Pixel

The two images on the left are the original images, while the images on the right are the images after applying the optimal transport map, T. The middle images are linear interpolations, $\mu_t = ((1-t)\mathrm{Id} + tT)_{\#}\mu$. We can see that the image 1 inbetween looks like fall and the image 2 transition looks like a picture from summer or spring. Thus color transfer is able to swap color pallets while maintaining the structure of the image.

2.2 Unequal Mass Discrete Measures

With equally distributed mass, it is simple to show that the Monge's formulation exists. As noted previously, if the masses of two discrete measures are unequally distributed, then it is

possible for no transport maps to exist. Thus to approach this problem, we need to consider the Kantorovich's formulation instead of the Monge's formulation.

Section 2.3 of [2] shows that the set of transport plans can be described as $\Pi(\mu,\nu) = \{T \in \mathbb{R}_+^{n\times m} : T\mathbb{1}_m = \mu,\ T^T\mathbb{1}_n = \nu\}$ and the Kantorovich's formulation becomes:

$$\min_{T\in\Pi} \langle C, T\rangle := \sum_{i,j} C_{ij}T_{ij} \tag{5}$$

One problem where this formulation is useful would be resource optimization. Suppose we have a resource, such as energy, stored at locations x_i with α_i amount of that resource and we have demand for the resource at locations y_i with the demand of β_i amounts. Then both of these probability densities are unequally distributed discrete measures, so this formulation finds the optimal plan to send resources from the stored locations to the places that demand the resources.

3 Wasserstein Distance

3.1 Introduction to the Wasserstein Distance

The Wasserstein Distance is a metric between probability measures using optimal transport. Prior to the Wasserstein Distance, there were no metrics on probability measures. There were, however, some divergences such as the KL-divergence.

$$D_{KL}(P||Q) = \int p(x)\log\left(\frac{p(x)}{q(x)}\right)dx \tag{6}$$

Since these divergences are not metrics, there are always shortcomings in using them. The main problems for the KL-divergence is that it is not symmetrical, i.e. $D_{KL}(P||Q) \neq D_{KL}(Q||P)$ and if P and Q do not have the same support then $D_{KL} = +\infty$. [4]

Before the definition of the Wasserstein Distance, we must define $P_p(X)$

Definition 5. ***Space of Probability Measures with Bounded p-th Moment*** *The space of probability measures with bounded p-th moment is denoted by $P_p(X)$ and defined by*
$P_p(X) := \{\mu \in P(X) : \int_X |x|^p d\mu(x) < +\infty\}$

This leads us to the definition of the Wasserstein Distance.

Definition 6. ***Wasserstein Distance*** *Given two probability measures in $P_p(X)$, μ and ν, then the Wasserstein Distance is as follows:*

$$d_{W^p}(\mu,\nu) = \min_{\pi\in\Pi(\mu,\nu)} \left(\int_{X\times X} |x-y|^p d\pi(x,y)\right)^{\frac{1}{p}}$$

The condition of $\mu,\nu \in P_p(X)$ is sufficient for the Wasserstein Metric to be finite. To see this, note that:

$$\begin{aligned} d_{W^p}(\mu,\nu) &= \min_{\pi\in\Pi(\mu,\nu)} \left(\int_{X\times X} |x-y|^p d\pi(x,y)\right)^{\frac{1}{p}} \\ &\leq \min_{\pi\in\Pi(\mu,\nu)} \left(\int_{X\times X} |x|^p + |y|^p d\pi(x,y)\right)^{\frac{1}{p}} \\ &\leq \left(\int_X |x|^p d\mu(x) + \int_X |y|^p d\nu(y)\right)^{\frac{1}{p}} < \infty \end{aligned} \tag{7}$$

We will now show that the Wasserstein distance is a metric.

Theorem 1. *The Wasserstein Distance, $d_{W^p} : P^p(X)\times P^p(X) \to [0,\infty)$, is a metric.*

Proof First note that $|x-y|^p \geq 0\ \ \forall x,y \in X$, thus $d_{W^p}(\mu,\nu)\ \ \forall \mu,\nu \in P^p(X)$. Now, we must show that d_{W^p} satisfies the 4 properties of being a metric.

1. $d_{W^p}(\mu,\nu) = d_{W^p}(\nu,\mu) \ \ \forall \mu,\nu \in P^p(X)$:
First, notice $c(x,y) = |x-y|^p = |y-x|^p = c(y,x)$ and $\pi \in \Pi(\mu,\nu) \implies S_{\#}\pi \in \Pi(\nu,\mu)$, where $S(x,y) = (y,x)$. Let $\pi \in \Pi(\mu,\nu)$ then we have,

$$\begin{aligned}\left(\int_{X\times X} |x-y|^p d\pi(x,y)\right)^{\frac{1}{p}} &= \left(\int_{X\times X} |y-x|^p d(S_{\#}\pi)(y,x)\right)^{\frac{1}{p}} \\ &\leq \min_{\pi\in\Pi(\nu,\mu)} \left(\int_{X\times X} |x-y|^p d\pi(y,x)\right)^{\frac{1}{p}} = d_{W^p}(\nu,\mu)\end{aligned} \tag{8}$$

Since π was arbitrary, we have that $d_{W^p}(\mu,\nu) \leq d_{W^p}(\nu,\mu)$. Similarly we have,

$$\begin{aligned}\left(\int_{X\times X} |y-x|^p d\pi(y,x)\right)^{\frac{1}{p}} &= \left(\int_{X\times X} |x-y|^p d(S_{\#}\pi)(x,y)\right)^{\frac{1}{p}} \\ &\leq \min_{\pi\in\Pi(\nu,\mu)} \left(\int_{X\times X} |x-y|^p d\pi(x,y)\right)^{\frac{1}{p}} = d_{W^p}(\mu,\nu)\end{aligned} \tag{9}$$

So we have, $d_{W^p}(\mu,\nu) \leq d_{W^p}(\nu,\mu)$ and $d_{W^p}(\nu,\mu) \leq d_{W^p}(\mu,\nu)$, which implies that $d_{W^p}(\mu,\nu) = d_{W^p}(\nu,\mu)$

2. $\mu = \nu \implies d_{W^p}(\mu,\nu) = 0$
Suppose that $\mu = \nu$, then we let $\pi(x,y) = \delta_x(y)\mu(x)$, then we have that,

$$0 \leq d_{W^p}(\mu,\nu) \leq \left(\int_{X\times X} |x-y|^p d\pi(x,y)]\right)^{\frac{1}{p}} = 0 \tag{10}$$

3. $d_{W^p}(\mu,\nu) = 0 \implies \mu = \nu$
Suppose that $d_{W^p}(\mu,\nu) = 0$, then by above, $\exists\pi$ such that $x = y$ π-almost everywhere. Thus, for any function f, we have that,

$$\int_X f(x)d\mu(x) = \int_{X\times X} f(x)d\pi(x,y) = \int_{X\times X} f(y)d\pi(x,y) = \int_{X\times X} f(y)d\nu(y) \tag{11}$$

Since this is true for any function, we have that $\mu = \nu$.

4. The triangle inequality
In order to prove the triangle inequality, we will need to cite an intermediate lemma. Lemma 5.5 from [3] states that given measures, $\mu,\nu,\omega \in P(X)$ and transport plans $\pi_1 \in \Pi(\mu,\nu)$ and $\pi_2 \in \Pi(\nu,\omega)$, then there exists a measure $\gamma \in P(X\times X\times X)$ such that $P^1_{\#}\gamma = \pi_1$ and $P^2_{\#}\gamma = \pi_2$, where $P^1(x,y,z) = (x,y)$ and $P^2(x,y,z) = (y,z)$ are projections onto their two respective variables. Now to prove the triangle inequality for the Wasserstein distance. Suppose $\mu,\nu,\omega \in P_p(X)$ and $\pi_1 \in \Pi(\mu,\nu)$ and $\pi_2 \in \Pi(\nu,\omega)$ be optimal transport plans. Let $\gamma \in P(X\times X\times X)$ be the γ from the above lemma. Then, $P^1_{\#}\gamma = \pi_1$ and $P^2_{\#}\gamma = \pi_2$. We let $\pi_3 = P^3_{\#}\gamma = (P^2 \circ P^1)_{\#}\gamma$. Then for any $A \subseteq X$, we have:

$$\begin{aligned}\pi_3(A\times X) &= P^3_{\#}\gamma(A\times X) \\ &= \gamma\left(\{(x,y,z) : P^3(x,y,z) = (x,z) \in A\times X\}\right) \\ &= \gamma\left(\{(x,y,z : x \in A\}\right) \\ &= \gamma(A\times X\times X) \\ &= \pi_3(A\times X) \\ &= \mu(A)\end{aligned} \tag{12}$$

Similarly, we find that $\pi_3(X\times B) = \omega(B) \ \ \forall B \in X$. Thus π_3 has the proper marginals,

which makes it a proper transport plan. Now we can finally prove the triangle inequality,

$$\begin{aligned} d_{W^p}(\mu,\omega) &\leq \left(\int_{X\times X} |x-z|^p d\pi_3(x,z)\right)^{\frac{1}{p}} \\ &= \left(\int_{X\times X} |x-z|^p d\gamma(x,y,z)\right)^{\frac{1}{p}} \\ &\leq \left(\int_{X\times X} |x-y|^p + |y-z|^p d\gamma(x,y,z)\right)^{\frac{1}{p}} \\ &= \left(\int_{X\times X} |x-y|^p d\gamma(x,y,z)\right)^{\frac{1}{p}} + \left(\int_{X\times X} |y-z|^p d\gamma(x,y,z)\right)^{\frac{1}{p}} \\ &= d_{W^p}(\mu,\nu) + d_{W^p}(\nu,\omega) \end{aligned} \tag{13}$$

Thus we have proved that the Wasserstein distance is a metric.

3.2 Applications to Machine Learning

Now that we have introduced the Wasserstein distance and laid out a couple import results, we will discuss how and why the Wasserstein distance is used in machine learning.

One of the most popular uses of the Wasserstein distance recently is the use of the metric in generative adversarial networks, GANs. GANs are a generative neural network model with the aim of generating new data, typically images, based off a set of training data. The typical way that a GAN model does this is by maximizing the problem $\max_{\theta\in\mathbb{R}^d} \frac{1}{m}\sum_{i=1}^{m} \log P_\theta(x_i)$, where θ is the set of parameters in our model, P_θ is a density function based on θ, and x_i is the training data. This formulation of GANs leads to minimizing the KL-divergence, but as stated earlier, there are many undesirable traits of the KL-divergence. One reason to introduce the Wasserstein distance into the GAN formulation is because [1] shows in theorem 1 that if μ_r is a fixed distribution over X, g a function from $Z\times\mathbb{R}^d \to X$ denoted $g_\theta(z)$, with Z a random variable, and ν_θ the distribution of $g_\theta(Z)$. Then if g is continuous in θ, so is $d_{W^p}(\mu,\nu_\theta)$ and if g is locally Lipschitz, the $d_{W^p}(\mu,\nu_\theta)$ is continuous everywhere and differentiable almost everywhere. Thus the Wasserstein distance is a continuous and almost differentiable loss function, which improves stability of learning and reduces mode collapse.

Another use of the Wasserstein distance is the use in network adversarial robustness for image classification. To define adversarial robustness in broad terms, let C be the set of classes the network will identify, let $f:[0,255]^n \to C$ be the neural network, then we have an image of our classes, $x_i \in [0,255]^n$ such that the proper class of x_i is c_j, then an adversarial attack is the optimization problem, $\min|\epsilon|$ subject to $f(x_i+\epsilon_i)\neq c_j$. In plain terms, this equates to finding the smallest amount of noise in order to fool the neural network into classifying an image into the wrong classification. These attacks can be potentially dangerous if they are used against networks, such as the networks behind self driving cars. Thus, it is important that we defend against them. Since the Wasserstein distance is continuous and almost everywhere differentiable, $x_i+\epsilon$, should be close in distance to the original image when using the Wasserstein distance. Thus, [5] shows that using the Wasserstein distance as the cost function leads to a more robust network.

References

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